

Minimax Chess

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Introduction

- Our main goal was to create a system that not only plays chess using adversarial search, but also demonstrates core course ideas such as heuristic evaluation, zero-sum modeling, and intelligent decision-making under uncertainty.
- Another goal was to enhance the user's experience by adding custom pieces to the board, such as the super_knight (a modified knight that can go in a perfect L, which is 2 moves horizontal and 2 moves vertical)
- Lastly, we wanted to add an AI referee that can analyze the state of the board, for better analysis.

Methods Summary

Base Chess Algorithm

- Uses iterative deepening with a **minimax** search limited to **depth 4** due to resource constraints.

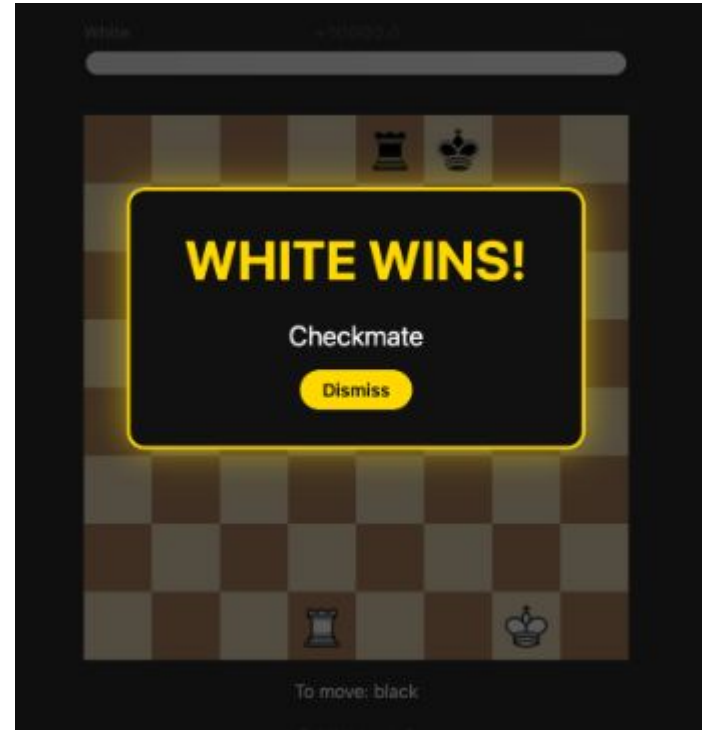
LLM Referee

- Every **10 moves** (5 per player), the LLM analyzes the current game state using:
 - Minimax evaluation score
 - Board pieces and their relative positions
- Provides users with periodic strategic assessments.

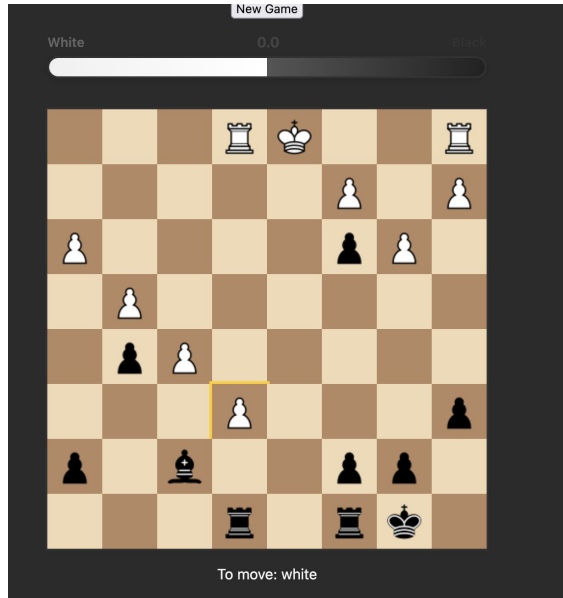
Custom Piece: Super-Knight

- Moves like a knight but **2 squares horizontally and 2 vertically**.
- Assigned a value of **8 points** (5 more than a traditional knight).

Finding a Mate in One - Ayush Pai, Richard Tan, Luke Allen, Mihika Gupta, Nat Heddaeus



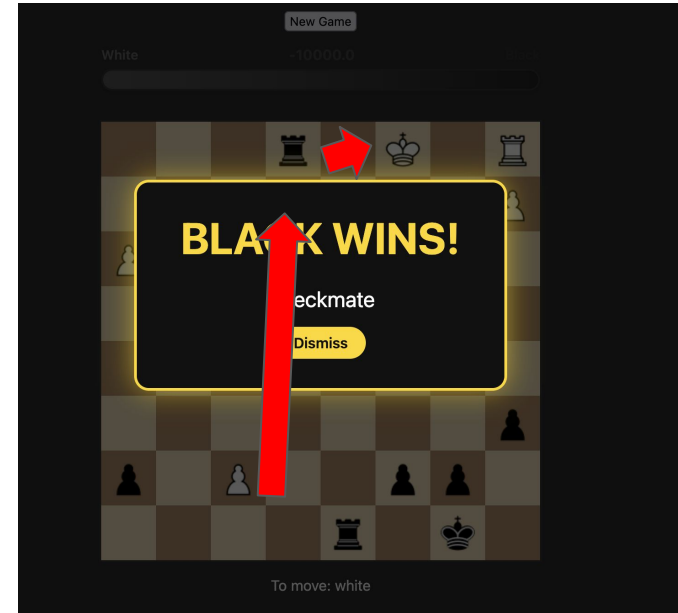
Finding a Mate in Two



White pawn moves from d3 to c2, black rook moves from f1 to e1



White king moves from e8 to f8, black rook moves from d1 to d8 to claim checkmate





To move: white

Total moves: 10

Referee Commentary

It's White who currently holds the advantage with a score of 1.0, thanks to their strong central pawn chain and control of key squares on the board. However, Black is fighting back with a cleverly placed rook on a8, which could potentially threaten White's position if they don't take action soon. The real question now is whether White can maintain their advantage while navigating the complex web of pawns and pieces on the board.

☒ Play vs engine

New Game



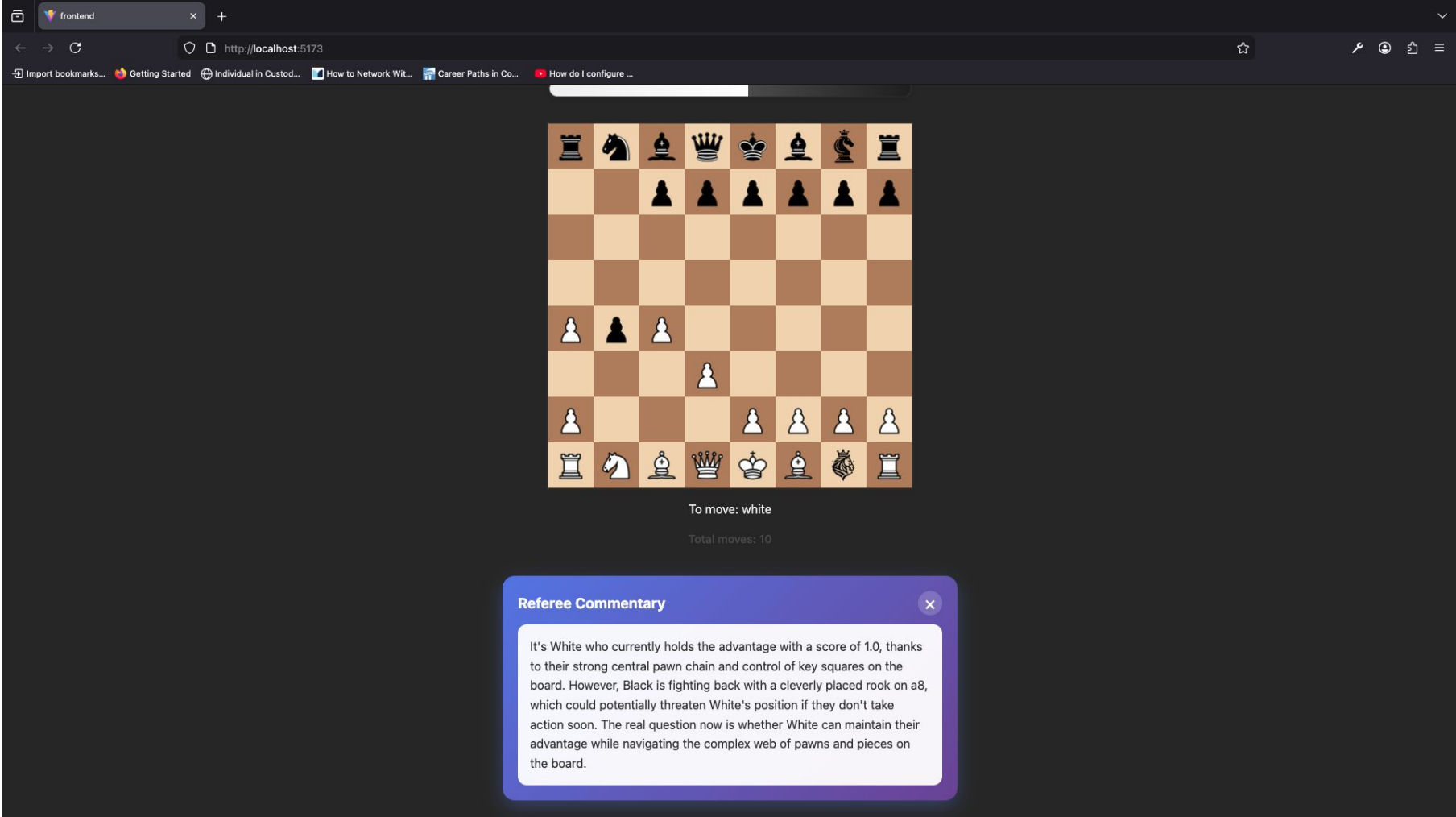
Black

White

6.0

To move: white

Total moves: 46

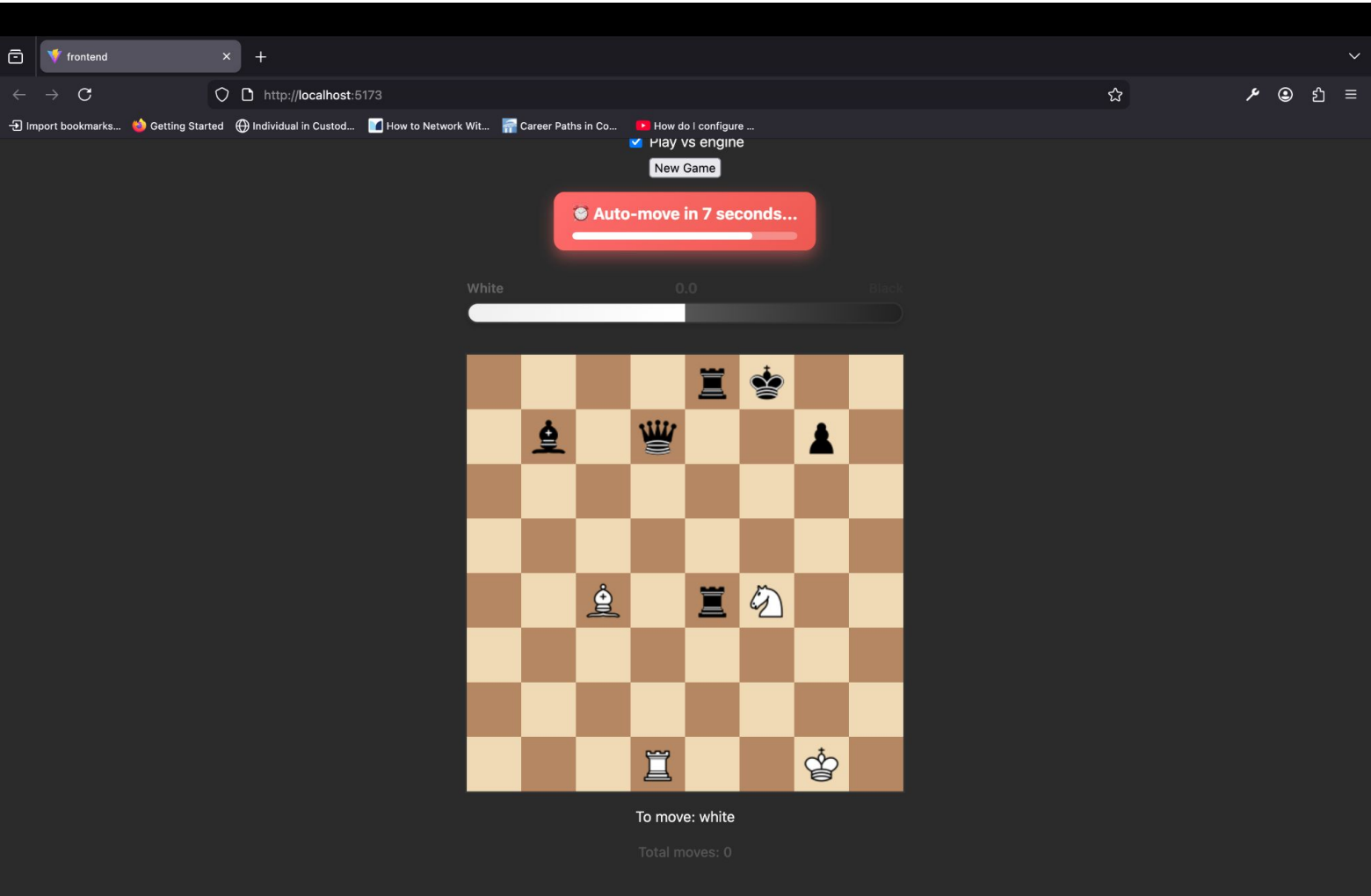


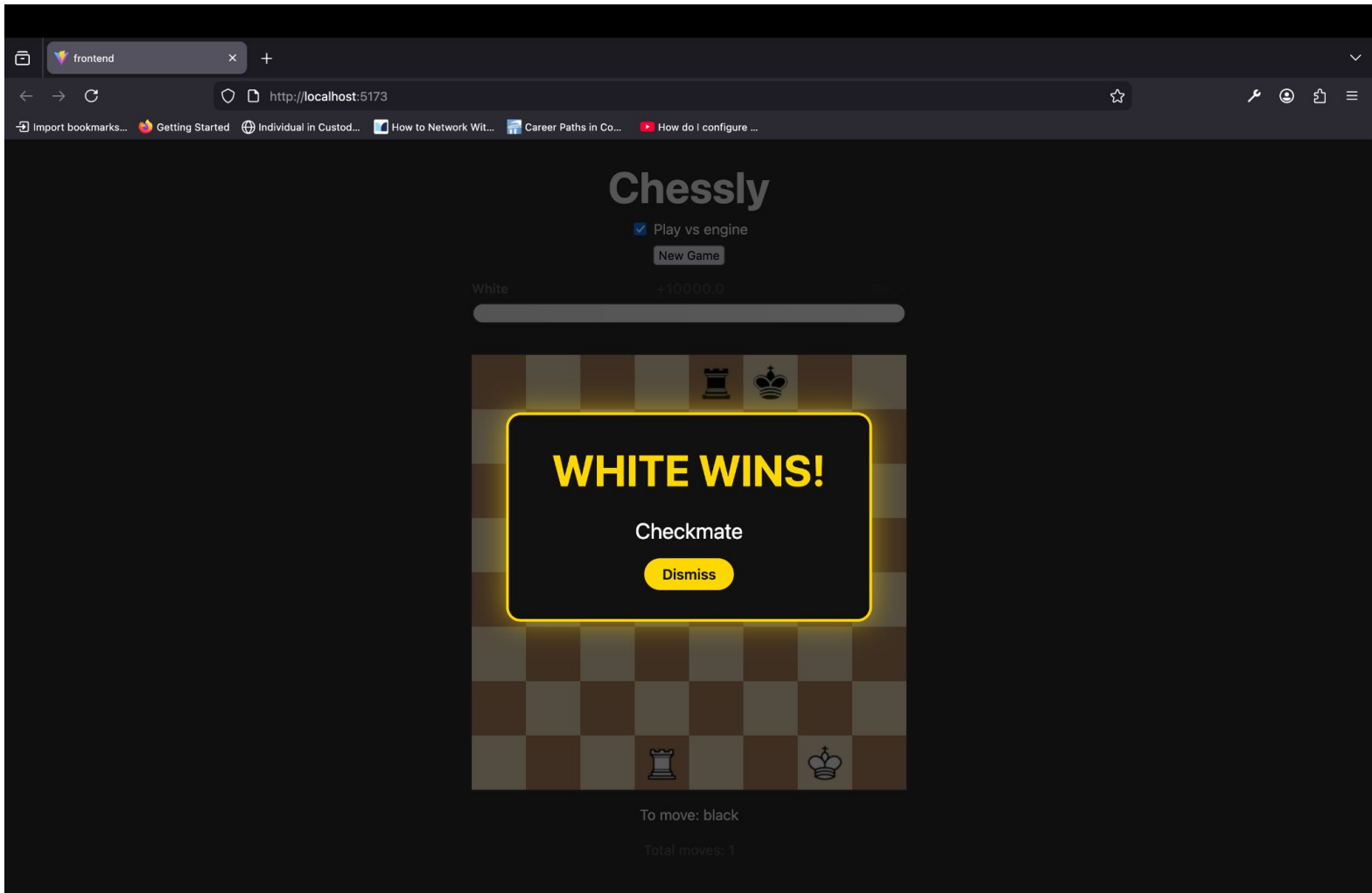
To move: white

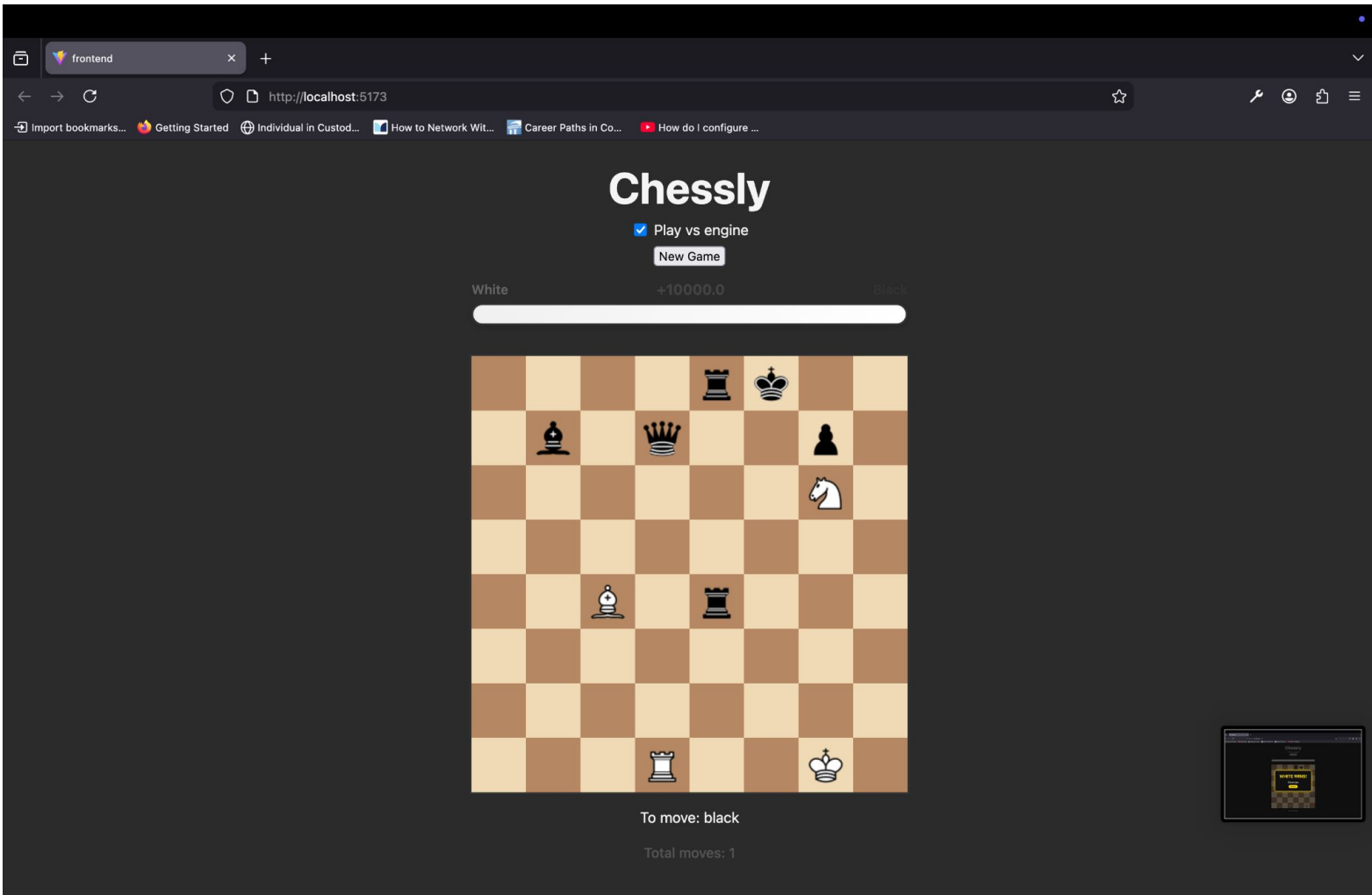
Total moves: 10

Referee Commentary

It's White who currently holds the advantage with a score of 1.0, thanks to their strong central pawn chain and control of key squares on the board. However, Black is fighting back with a cleverly placed rook on a8, which could potentially threaten White's position if they don't take action soon. The real question now is whether White can maintain their advantage while navigating the complex web of pawns and pieces on the board.







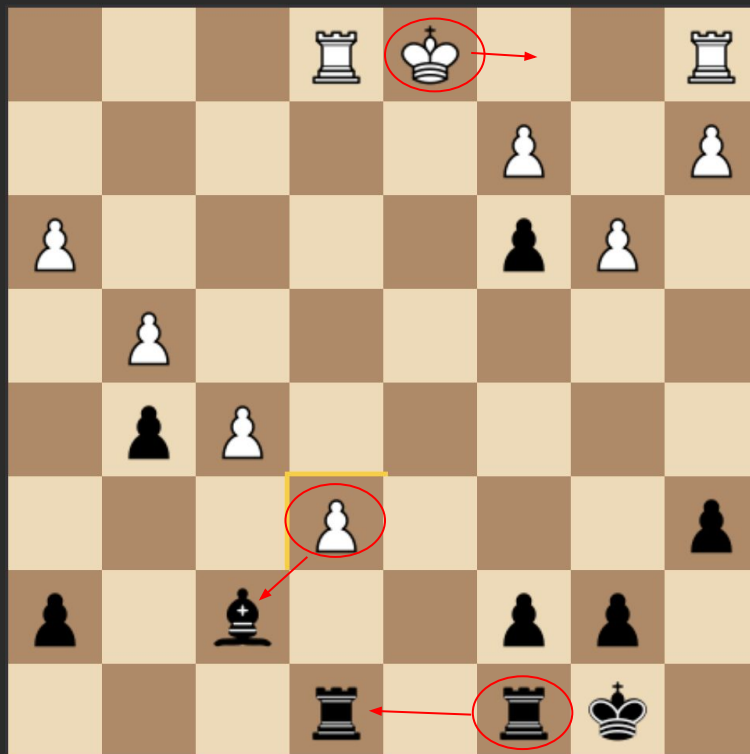
2 - mate

New Game

White

0.0

Black



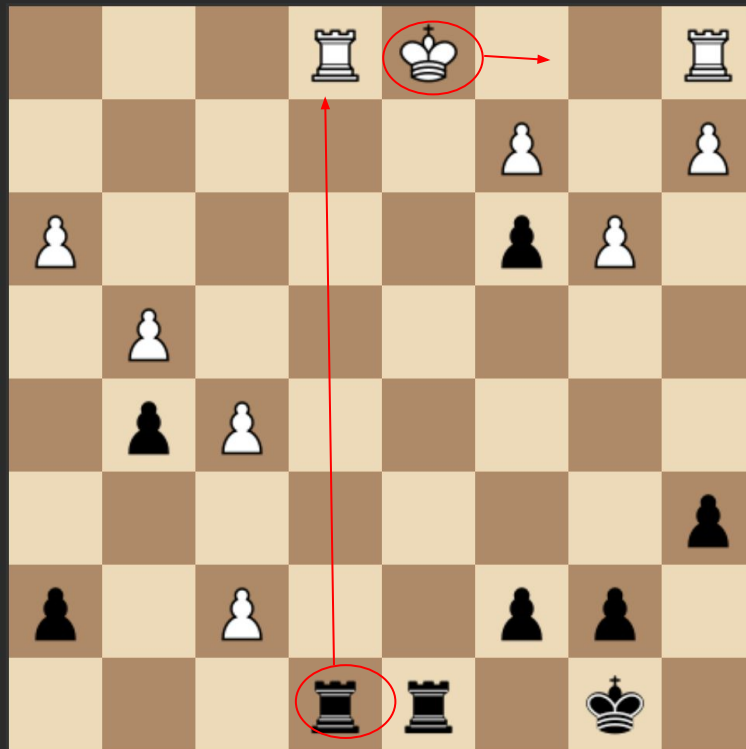
To move: white

New Game

White

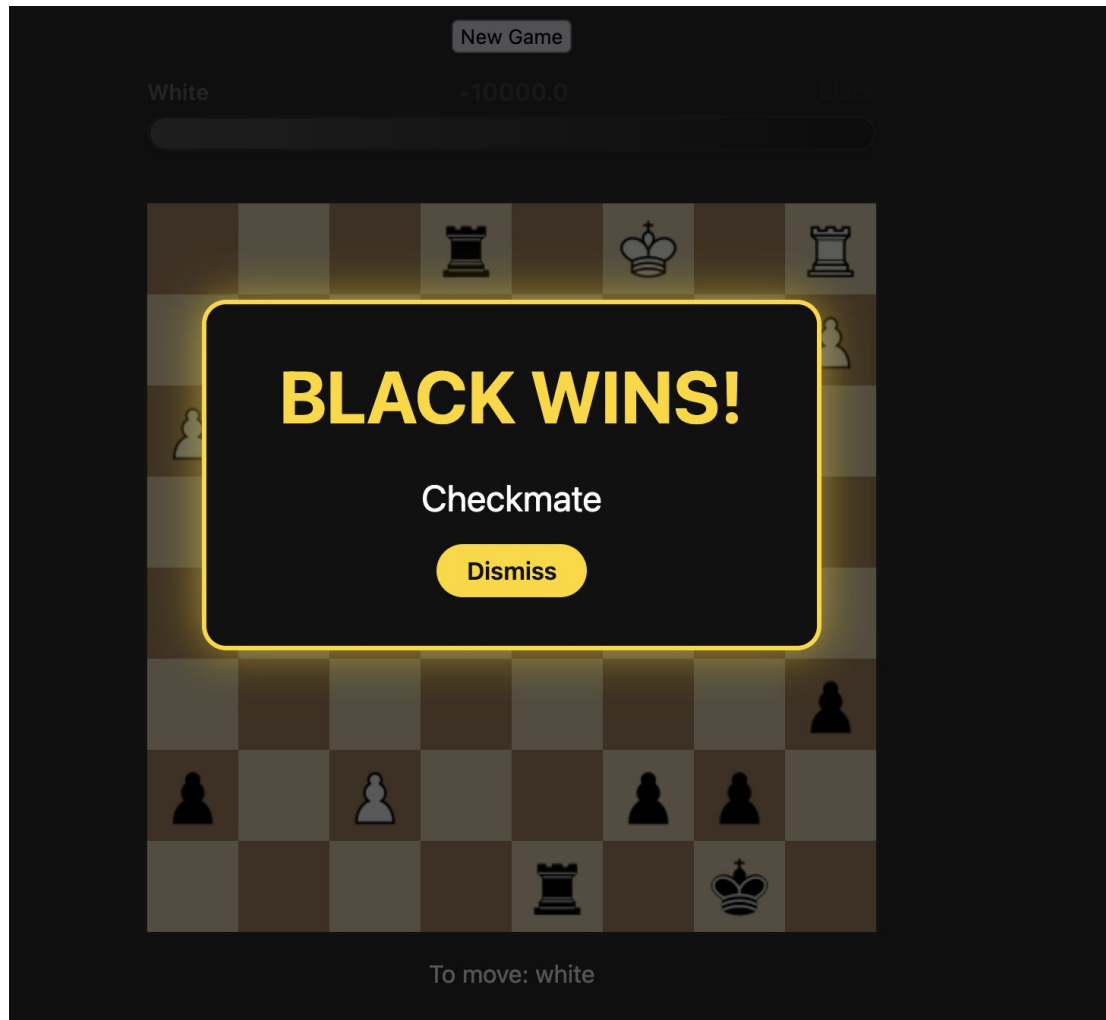
+1.0

Black



To move: white

White king moves from e8 to f8, black rook moves from d1 to d8 to claim checkmate





To move: black

